



NeuroBabyLM: A Benchmark for Brain Alignment in Language Models Trained on Developmentally-Plausible Datasets

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Introduction

NeuroBabyLM == BabyLM + Brain Alignment Score

- Current language models are trained on massive, developmentally implausible amounts of data.
- BabyLM Challenge¹ constrains training to fixed dataset sizes comparable to child language input.
- NeurobabyLM incorporates brain alignment scores, evaluating models at both behavioral and computational levels.
- Brain alignment scores, measured via encoding models, assess similarity between model and human processing of linguistic stimuli.



BabyLM Challenge¹

- Dataset

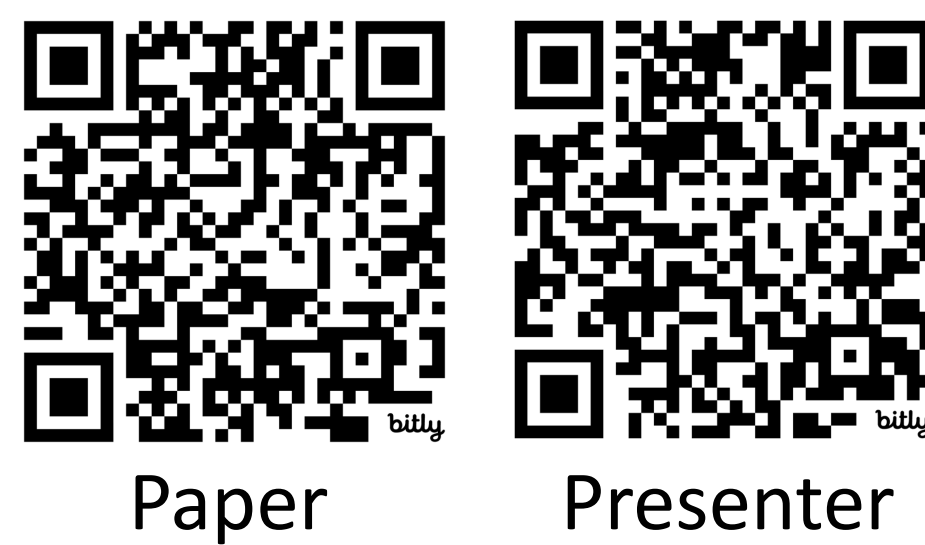
Simplified language, such as child-directed speech, children's storybooks, and simple Wikipedia.

Dataset	Description	# Words (multimodal)	# Words (strict)	# Images
Localized Narratives ^a	Image Caption	27M	–	0.6M
Conceptual Captions 3M ^b	Image Caption	23M	–	2.3M
CHILDES ^c	Child-directed speech	14.5M	29M	–
British National Corpus (BNC), dialogue portion ^d	Dialogue	4M	8M	–
Project Gutenberg (children's stories) ^e	Written English	13M	26M	–
OpenSubtitles ^f	Movie subtitles	10M	20M	–
Simple English Wikipedia ^g	Written Simple English	7.5M	15M	–
Switchboard Dialog Act Corpus ^h	Dialogue	0.5M	1M	–
Total	–	100M	100M	2.9M

Table 1: Datasets for the multimodal and strict tracks of the 2nd BabyLM Challenge. Word counts are approximate and subject to slight changes. ^aPont-Tuset et al. (2020a) ^bSharma et al. (2018a) ^cMacWhinney (2000) ^dConsortium (2007) ^eGerlach and Font-Clos (2018) ^fLison and Tiedemann (2016a) ^g<https://dumps.wikimedia.org/simplewiki/> ^hStolcke et al. (2000)

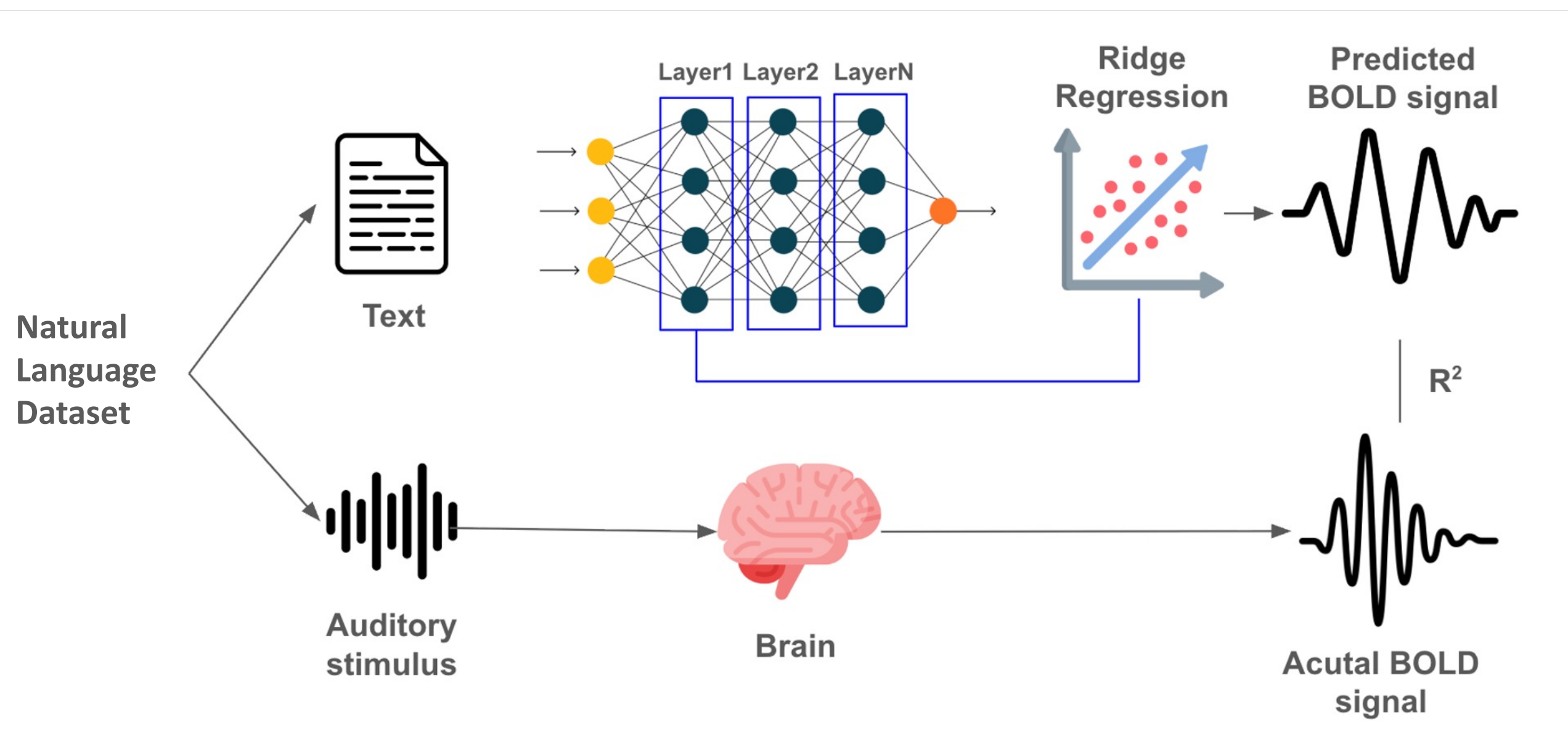
- Track

- Strict(100M), Strict-Small(10M)
- Multimodal
- Interaction



Methods

Encoding model



Encoding Dataset²

- 3T fMRI / 3 subjects / ~20hrs of English podcast
- Train set : 95 stories, 20 scanning sessions (33,000 timepoints)
- Test set : three held-out stories

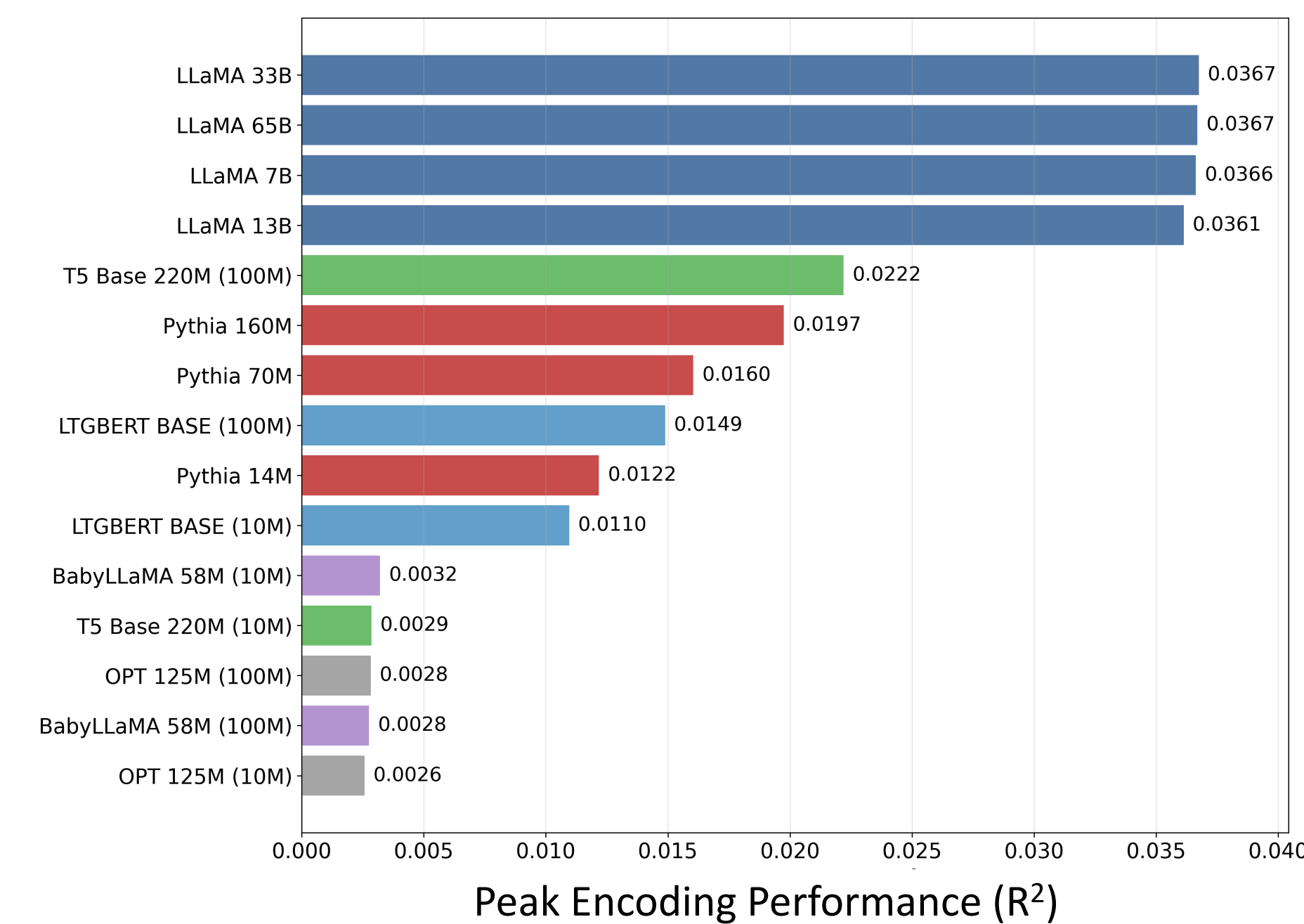
Tested Models

Table 1: Summary of Tested Language Models and Corresponding Analyzed Layers

Model Family	Model Name	Layers	Tested Layers	Dataset Size
babylm ¹	babylama-58m-10m	16	1, 3, 5, 7, 9, 11, 13, 15	10M words
	babylama-58m-100m	16	0, 4, 8, 12, 15	100M words
	opt-125m-strict	12	0, 4, 8, 11	10M words
	opt-125m-strict-small	12	0, 4, 8, 11	100M words
	ltgbert-10m	12	0, 4, 8, 11	10M words
	ltgbert-100m	12	0, 4, 8, 11	100M words
	t5-base-strict-small	12	0, 4, 8, 11	10M words
EleutherAI ⁵	pythia-14m	6	0, 1, 2, 3, 4, 5	300B tokens (est. 225B words)
	pythia-70m	6	0, 1, 2, 3, 4, 5	
	pythia-160m	12	0, 4, 8, 11	
	pythia-1b	16	-	
	pythia-6.9b	32	-	
	pythia-6.9b	32	-	
huggyllama ⁶	llama-7b	32	every second layer	1.4 T tokens (est. 1.05T words)
	llama-13b	32	every second layer	
	llama-33b	32	every second layer	
	llama-65b	32	every second layer	

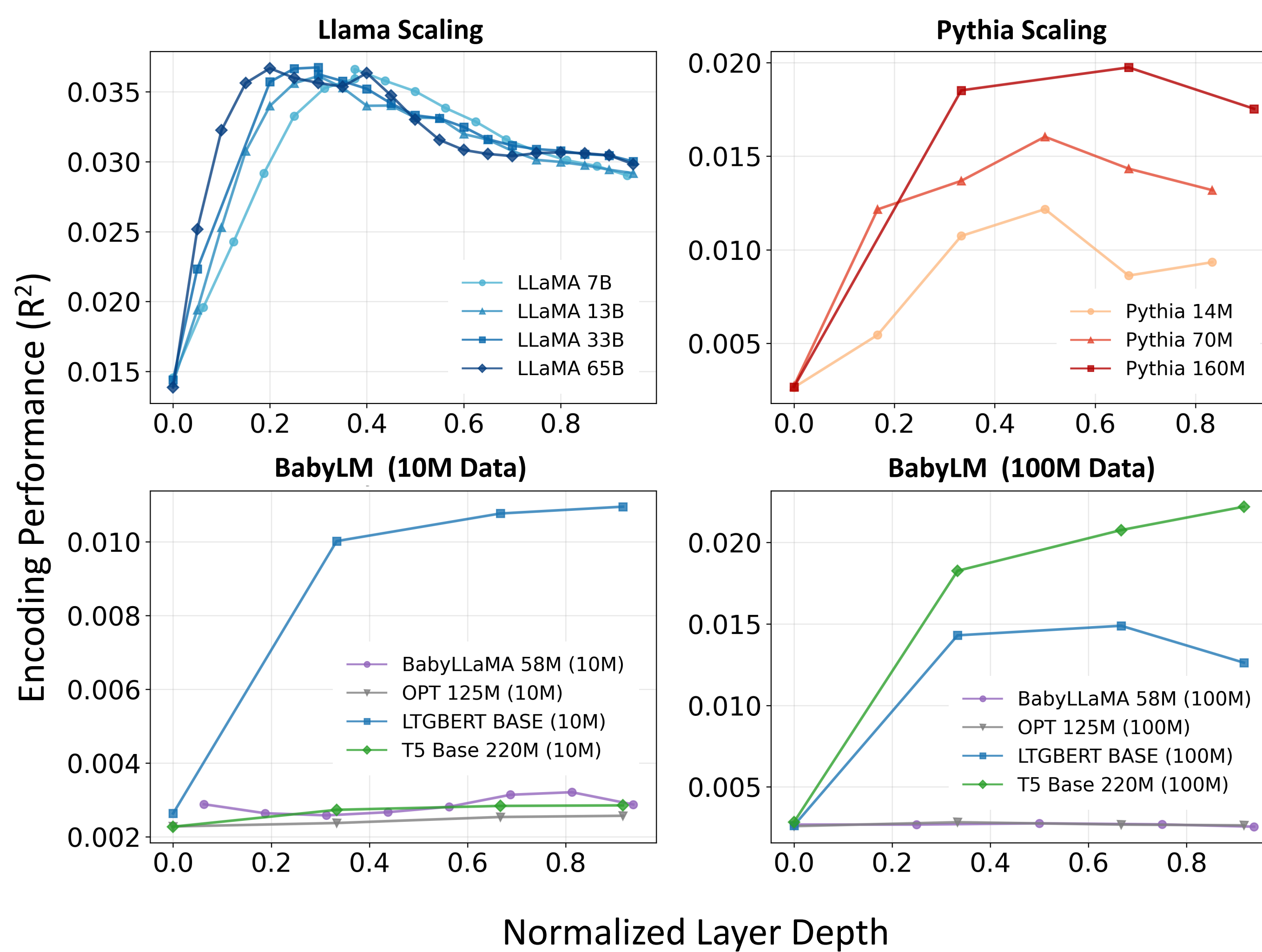
Results

- Overall performance

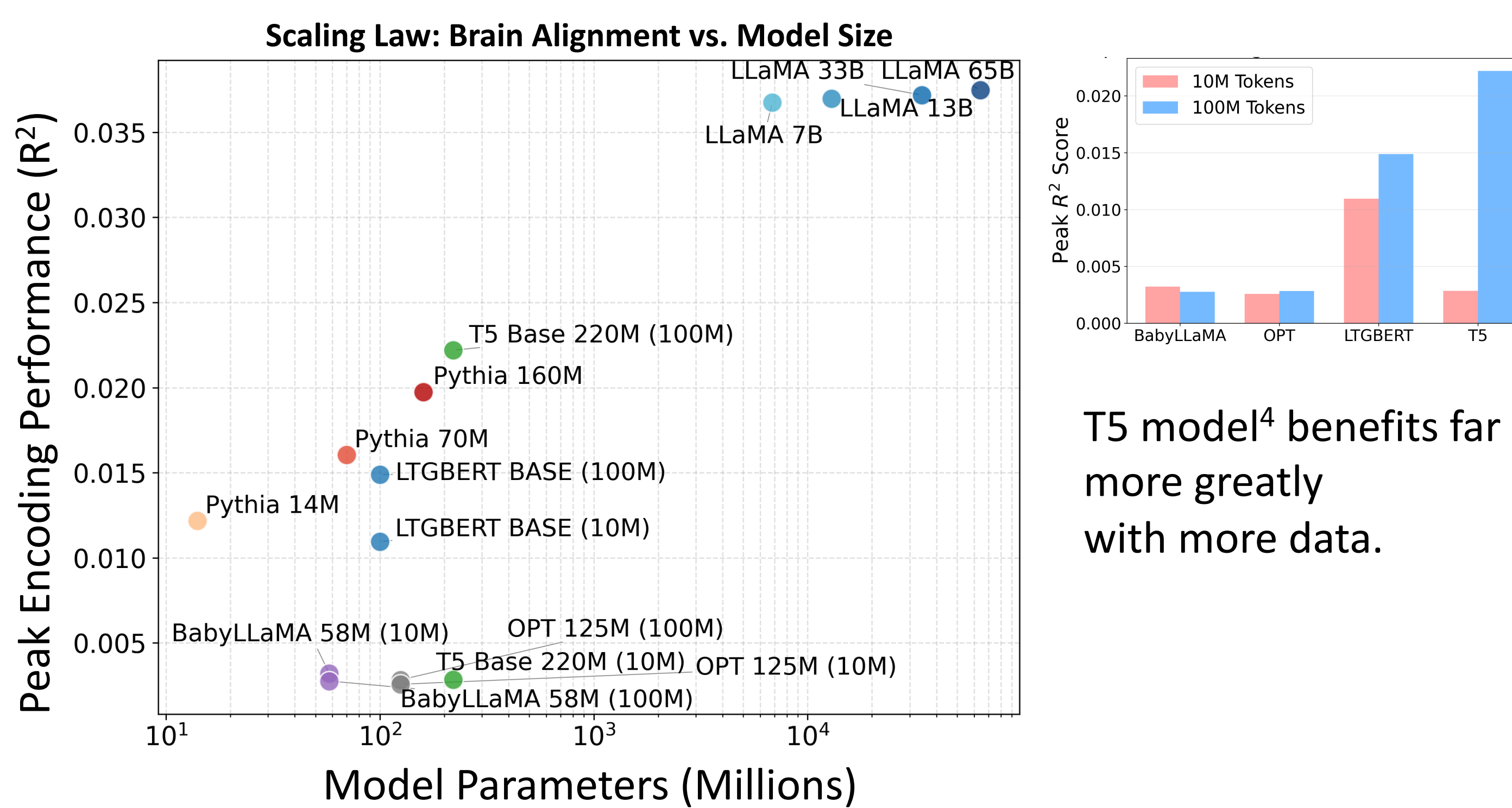


BabyLMs^{1,3,4} are still far away from the ceiling⁶.

T5⁴ Base BabyLM beats similarly sized Pythia⁵ 160M with x2000 less data. (100M vs 225B)



- Larger model achieve higher brain alignment score.
- Models trained on bigger data (10M → 100M words) show higher brain alignment score.



T5 model⁴ benefits far more greatly with more data.

Discussion

Findings

- Larger models and more training data improve performance.
- Model architecture independently contributes to brain alignment scores.

Future Works

- Analyze BabyLM benchmark score (e.g. BLIMP/MMLU) with brain alignment score, Test on +50 BabyLM models
- Expand to different modalities (e.g. ECoG, MEG)

Reference

- <https://babylm.github.io/>
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